

# Module 09 — Explainability Layer

models/explainability.py • 397 lines

## In one sentence

Answers 'why was this recommended?' - worth 10 marks in the rubric.

## 1. What this module does

A recommender that cannot explain itself is hard to deploy. A merchandising team will not hand over homepage space to something they cannot question when it does something odd, and in a growing number of markets there are legal transparency requirements too.

This module produces the three explanation types the business case names: why an item was recommended, which similar customers or products support it, and what content similarity justifies it.

The design rule throughout: **every explanation is built from the actual scores that produced the ranking**, never from a template. A plausible-sounding sentence that does not match the real reason is worse than saying nothing.

## 2. Concepts covered

Every idea this module uses, in plain terms.

| Concept                     | In simple terms                           | Where it is used here             |
|-----------------------------|-------------------------------------------|-----------------------------------|
| Explainable AI (XAI)        | Making a model's decisions understandable | The whole module                  |
| Feature attribution         | How much did each input contribute?       | <code>attribute_signals()</code>  |
| Contribution share          | weight x score, as a percentage           | The attribution table             |
| Evidence-based explanation  | Point at real data, not a story           | Uses the customer's own purchases |
| Counterfactual honesty      | Say when you do not know                  | Empty-evidence handling           |
| Term-level attribution      | Which exact words caused the match        | <code>top_shared_terms()</code>   |
| Transparency and governance | Being auditable                           | Why this module exists            |

## 3. How it works, step by step

- 1. Attribution.** Multiply each signal's score by its weight, then express each as a share of the total. This is what says '51% of this came from item-based CF'.
- 2. Build the sentence.** Add a clause only when the underlying number is actually high. If a signal did not fire, its clause is not written.
- 3. Find supporting evidence** from the customer's own past purchases - the most persuasive kind, because they can verify it themselves.
- 4. Check similar customers** - how many of the 50 nearest also rated this highly.
- 5. Explain the content link** by showing the shared category, brand, tags and the actual matching words.

**6. For cold-start customers**, produce a visibly different and honest explanation that says the recommendation is based on registration details.

## 4. Key functions

| Function                                    | What it does                                 |
|---------------------------------------------|----------------------------------------------|
| <code>attribute_signals()</code>            | Breaks a score into per-signal contributions |
| <code>dominant_signal()</code>              | Which signal drove this, and by how much     |
| <code>explain_recommendation()</code>       | The main human-readable sentence             |
| <code>explain_via_similar_items()</code>    | The customer's own comparable purchases      |
| <code>explain_via_similar_users()</code>    | How many similar customers liked it          |
| <code>render_similar_user_evidence()</code> | Phrases that evidence, including when empty  |
| <code>explain_content_similarity()</code>   | Shared terms, tags and attributes            |
| <code>build_full_explanation()</code>       | All three types as structured data           |

## 5. Inputs, outputs and how to run

|          | Detail                                                           |
|----------|------------------------------------------------------------------|
| Inputs   | A recommendation row plus the loaded recommender                 |
| Outputs  | Explanation text and structured evidence                         |
| Used by  | FastAPI /explain endpoint and the dashboard's Explainability tab |
| Run with | <code>python models/explainability.py</code>                     |

## 6. Key code

models/explainability.py — attribute\_signals()

```

47 | def attribute_signals(recommendation_row: pd.Series, weights: dict) -> pd.DataFrame:
48 |     """Decompose a hybrid score into each signal's actual contribution."""
49 |     # Report weight x score, not the raw score: a high score on a low-weight
50 |     # signal contributes little to the ranking.
51 |     rows = []
52 |
53 |     for column, weight_key in SIGNAL_WEIGHT_KEYS.items():
54 |         score = float(recommendation_row.get(column, 0.0) or 0.0)
55 |         weight = float(weights.get(weight_key, 0.0))
56 |         rows.append({
57 |             "signal": SIGNAL_LABELS[column],
58 |             "raw_score": score,
59 |             "weight": weight,
60 |             "contribution": score * weight,
61 |         })
62 |
63 |     frame = pd.DataFrame(rows)
64 |     total = frame["contribution"].sum()
65 |     frame["contribution_share"] = frame["contribution"] / total if total > 0 else 0.0
66 |
67 |     return frame.sort_values("contribution", ascending=False).reset_index(drop=True)

```

models/explainability.py — render\_similar\_user\_evidence()

```

212 | def render_similar_user_evidence(evidence: dict) -> str:
213 |     """Phrase the collaborative evidence, including when there is none."""
214 |     # Empty evidence is expected: the re-ranker promotes long-tail items,
215 |     # which are long-tail precisely because few people interacted with them.
216 |     considered = evidence.get("neighbours_considered", 0)
217 |     rated = evidence.get("neighbours_who_rated", 0)
218 |     liked = evidence.get("neighbours_who_liked", 0)
219 |     mean_rating = evidence.get("mean_rating")
220 |
221 |     if considered == 0:
222 |         return "No comparable customers available for this account yet."
223 |
224 |     if rated == 0:
225 |         return (
226 |             f"None of your {considered} closest customer matches have rated this item - "
227 |             f"it is a low-exposure product surfaced for catalogue variety, so the "
228 |             f"recommendation rests on content and model signals rather than crowd evidence."
229 |         )
230 |
231 |     return (
232 |         f"{liked} of the {rated} closest customer matches who rated this item scored it "
233 |         f"{RELEVANCE_RATING_THRESHOLD} or above (mean {mean_rating}), out of {considered} "
234 |         f"neighbours considered."
235 |     )

```

*This function exists to be honest about missing evidence rather than hide it.*

## 7. Things to remember

### Why report weight x score rather than the raw score

A signal can have a very high raw score but a low weight, in which case it barely influenced the ranking. Showing the raw score alone would credit it for a decision it did not make. Contribution share is the honest measure.

### An honest tension worth raising yourself in the viva

Similar-customer evidence is often empty. The reason is structural, not a bug: the re-ranker deliberately promotes under-exposed products, and a product is under-exposed precisely because few people have bought it. **So the diversity feature undermines the evidence available to justify its own output.** The system states this plainly instead of hiding it - telling a customer 'people like you loved this' about a product nobody comparable has ever rated would be misleading.

**When the item is not long-tail, the evidence is strong**

In the live demo, a recommendation with normal exposure reported '19 of the 21 closest customer matches rated this 4 or above, mean 4.48'. So the mechanism works well - it is specifically the promoted long-tail items where crowd evidence is thin, and the wording adapts.

## 8. Likely questions

**Q. What are the three explanation types?**

A. Why the item was recommended (with signal attribution), supporting evidence from similar customers and the customer's own purchases, and the content-similarity justification showing the actual matching words.

**Q. How do you avoid making up explanations?**

A. Every clause is conditional on a real measured number. If the content score is low, no clause about content is written. A cold-start customer gets a completely different sentence that says outright it is based on registration details, not purchase history.

**Q. Why does it sometimes say there is no evidence?**

A. Because sometimes there genuinely is not, and saying so is better than inventing something. That happens most often for long-tail products the diversity re-ranker has promoted, which by definition few people have bought.